MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5703WB

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, subminiature, medium Mu, 500 MHz

Outline	---	3-6 (EIA)
Base	---	Pinch press (5 leads)
Envelope	---	T3
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	3	4	5	6
Element	---	a	h	h	g	k

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	Ehk	Rk	Rg	Ib	Ic	Pp	Tt	Alt
Unit:	V	Vdc	Vdc	v	Ohms	Meg	mAdc	mAdc	W	°C	ft
Maximum:	6.9	200	---	200	---	1.2	15	5.5	1.62	220	(see note 6)
Minimum:	5.7	---	---	---	---	---	---	---	---	---	
TEST CONDITIONS:	6.3	120	0	0	220	---	---	---	---	---	---

GENERAL:

Qualification - Required

Reliable tube

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 1								
1256	Electrode current (1) (anode)	1		0.65	II	Ib	7.3	11.5	mAdc
1256	Electrode current (2) (anode)		Ec = -8.5 Vdc	0.65	II	Ib	---	50	µAdc
1266	Total grid current	1		0.65	II	Ic	0	-0.3	µAdc
1301	Heater current			0.65	II	If	190	210	mA
1306	Transconductance (1)			0.65	II	Sm	4,300	5,700	µmhos
1336	Heater-cathode leakage			0.65	II	Ihk	---	5.0	µAdc
1201	Short and discontinuity detection			0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes			2.5	S3	R	250	---	Meg
1031	High-frequency vibration		Rp = 10,000 ohms	2.5	I	Ep	---	10	mV
1236	Power oscillation		f = 500 MHz; Eb = 150 Vdc; Rg/Ib = 20 mAdc	---	---	Po	600	---	m.
1246	Audio frequency noise		Esig = 50 mVac; Ec = -4.5 Vdc; Rp = 0.01 Meg; Rg = 0.1 Meg; Rk = 0	2.5	I	EB	---	17	vu
1256	Electrode current (3) (anode)		Ec = -5 Vdc	2.5	I	Ib	20	---	µAdc
1266	Grid emission	3	Ef = 7.5 V; Ec = -10 Vdc	---	---	Ic	0	-0.4	µAdc
1306	Transconductance (2)		Ef = 5.7 V	2.5	I	ΔSm Ef	---	5	%
1316	Amplification factor			---	---	Mu	22.5	28.5	---
1331	Direct-interelectrode capacitance		No shield	6.5	Code E	Cgp Cin Cout	1.0 2.0 0.65	1.6 3.2 1.05	pF pF pF
1231	Peak emission	4	Ef = 6.0 V; eb (pulse) = 50 v; tp = 25 µs; prr = 200	---	---	Is	300	---	mA

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 2 - Continued								
1256	Electrode current (1) (anode) (tube operation time)		t = 20 seconds	4.0	S3	Ib	90	110	%
1031	Sweep-frequency vibration		Rp = 10,000 ohms; F = 30 to 1,000 Hz; 15 G; t = 3 minutes; positions X and Y only	4.0	S3	ep	---	50	mv (p-p)
1116	Lead fatigue			2.5	Code F	---	---	---	---
2126	Envelope strain			2.5	I	---	---	---	---
1041	Shock	5	450 G; Ehk = 100 Vdc	---	---	---	---	---	---
1031	Vibration-fatigue (1)			6.5	See note 2	---	---	---	---
1031	Vibration-fatigue (2)		10 G; t = 6 hours	6.5	See note 2	---	---	---	---
---	Shock and vibration-fatigue (1) and (2)-test end points:								
1031	High-frequency vibration			---	---	Ep	---	20	mVac
1336	Heater-cathode leakage			---	---	Ihk	---	10	µAdc
1306	Change in transconductance (1) of individual tubes			---	---	ΔSmt	---	10	%
1266	Total grid current			---	---	Ic	---	-1.0	µAdc
1105	Permanence of marking			---	---	---	---	---	---

Symbol	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 3						
1506	Heater-cycling life		Ef = 7.5 V; Ehk = 140 Vac; Ec = 0; Eb = 0	---	---	---	---
---	Heater-cycling life-test end point:						
1336	Heater-cathode leakage			Ihk	---	20	µAdc
1516	Stability life		Ehk = 200 Vac; Rg = 1.0 Meg; TA = room	---	---	---	---
---	Stability life-test end point:						
1306	Change in transconductance (1) of individual tubes			ΔSmt	---	10	%

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Method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 3 - Continued						
1501	Intermittent life	7	Ehk = 200 Vdc; Rg = 1.0 Meg; TA = room; TE = 220°C (min); Group E	---	---	---	---
---	Intermittent life-test end points (500 hours):						
1266	Total grid current			Ic	0	-0.6	µAdc
1301	Heater current			If	187	217	mA
1306	Change in transconductance (1) of individual tubes			ΔSm t	---	20	%
1306	Transconductance (2)			ΔSm Ef	---	15	%
1336	Heater-cathode leakage			Ihk	---	10	µAdc
1211	Insulation of electrodes			R	50	---	Meg
1306	Transconductance (1), average change			Avg ΔSm t	---	15	%
---	Intermittent life-test end points (1,000 hours):						
1266	Total grid current			Ic	0	-1.0	µAdc
1301	Heater current			If	177	223	mA
1306	Change in transconductance (1) of individual tubes			ΔSm t	---	30	%
1336	Heater-cathode leakage			Ihk	---	15	µAdc

NOTES:

1. This test shall be performed at the conclusion of the holding period.
2. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot has passed. MIL-STD-105, sample size code letter E, shall apply.
3. Prior to this test, tubes shall be preheated a minimum of 5 minutes at the conditions indicated below. The 3-minute test shall not be permitted. Test at specified conditions, except the value of Ec shall be -10 Vdc, within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid emission test.

Ef	Ec	Eb	Rk	Rg
V	Vdc	Vdc	Ohms	Meg
7.5	0	120	220	1.0

4. The pulse is essentially a square wave with 1.0-µs rise time and 0.8-µs fall. The pulse shall be applied to anode and grid tied together. Pulse emission shall be measured in terms of voltage developed across a 1.0-ohm resistor in the cathode circuit. Limit shall be tested as measured by the leading edge of a calibrated oscilloscope trace, the amplitude of the trailing edge of which shall vary by not more than 20 percent from the value of the leading edge. (See figure 1.)

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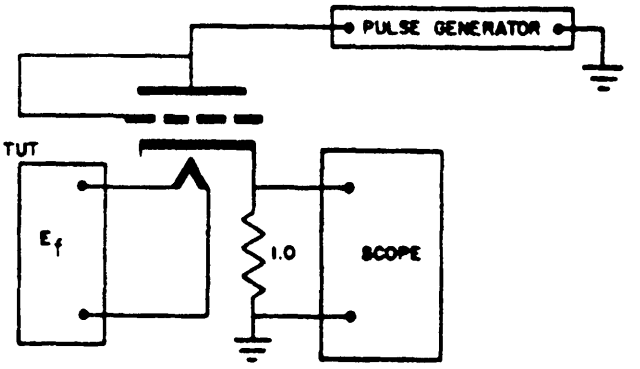


FIGURE 1. Pulse emission test circuit.

5. A grid resistor of 0.1 megohm shall be added; however, this resistor shall not be used when a thyratron-type short indicator is used.
6. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
7. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a TUT having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.
8. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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